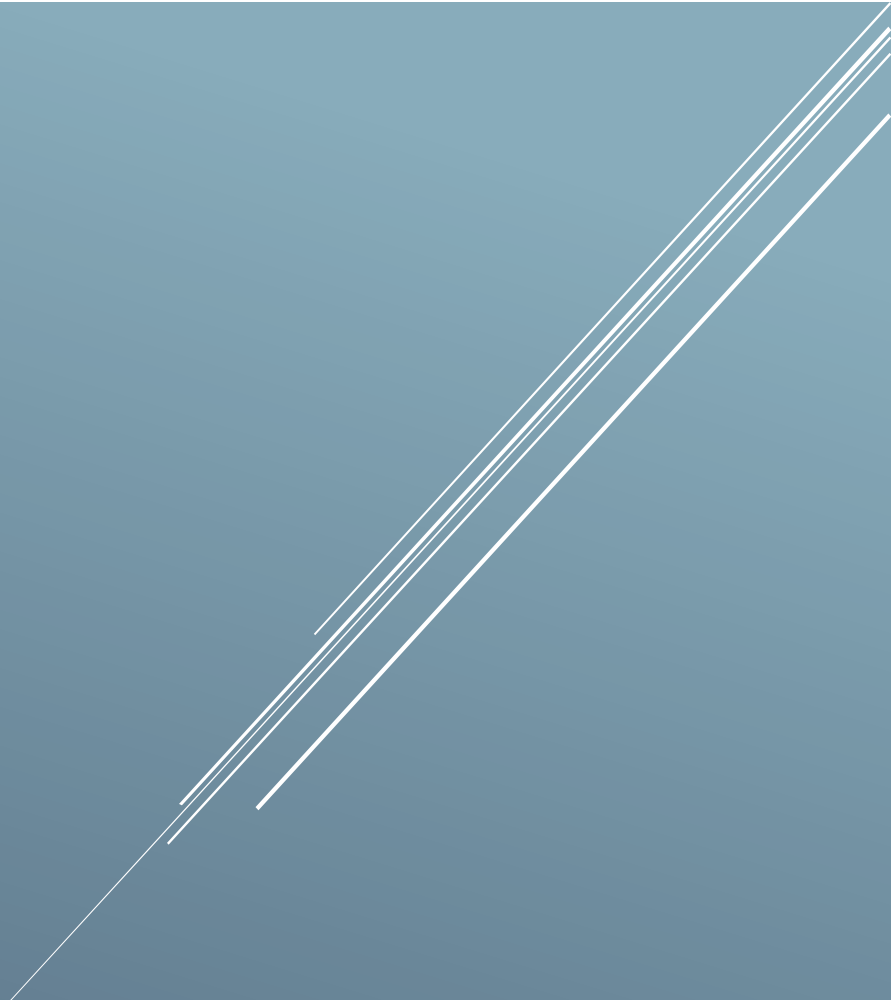




SDT621 FA2

Shatadi Nkadameng

Testing Objectives

System purpose, The Umoja Learning Management system (ULMS) is a cloud enabled system designed to manage student registrations, login authentication, course enrolment, marks capture and report generation

Testing goals:

- Verify that all system features work correctly
- Identify and document any defects found
- Ensure the system is ready for deployment
- Confirm that error messages display correctly
- Check that calculations like averages are correct

Scope of testing:

The following features will be tested:

- Student Registration
- Login Authentication
- Course enrolment
- Marks Capture
- Report Generation

Testing Types

Testing Types	What We Are Testing
Functional	Login validation
Input Validation	Mark entry (numbers only)
Integration	Course allocation
Performance	Report loading time
Usability	Navigation between screens
Security	Invalid login attempts

Testing Environment

Hardware:

- Personal Laptop

Software:

- Visual studio 2026

- Windows 11
- .net framework

Tools used:

- Visual studio (running and debugging code)
- Test data sets (fake student names and marks)

Task 2

Test Case ID	Description	Input	Expected Output	Actual Output	Status
TC001	Valid student login successful, dashboard open	Correct username and password	Login	Everything worked as expected	Pass
TC002	Invalid student login message: "Invalid credentials"	Wrong password	Error	Worked as expected	Pass
TC003	Empty login fields message: "Fields cannot be empty"	No username or password	Error	Worked as expected	Pass
TC004	Register new student saved successfully	Valid student details	Student	Saved student details	Pass
TC005	Register duplicate student "Student already exists"	Same student details twice	Error	Still saved the student details	Fail
TC006	Marks entry with numbers saved successfully	Enter 75 as a mark	Mark	Marks were saved successfully	Pass
TC007	Marks entry with letters "Please enter numbers only"	Enter "abc" as a mark	Error	Marks were not able to be saved with letters	Block
TC008	Course enrolment enrolled successfully	Enrol student in a course	Student	Enrols student successfully	Pass
TC009	Duplicate course enrolment "Already enrolled"	Enrol same students twice	Error	Still enrols student even though already enrolled	Fail
TC010	Report generation	Click reports button	Report	Generates report but slowly	Pass

	loads within 3 seconds				
--	---------------------------	--	--	--	--

Task 3

Valid Login

TC001

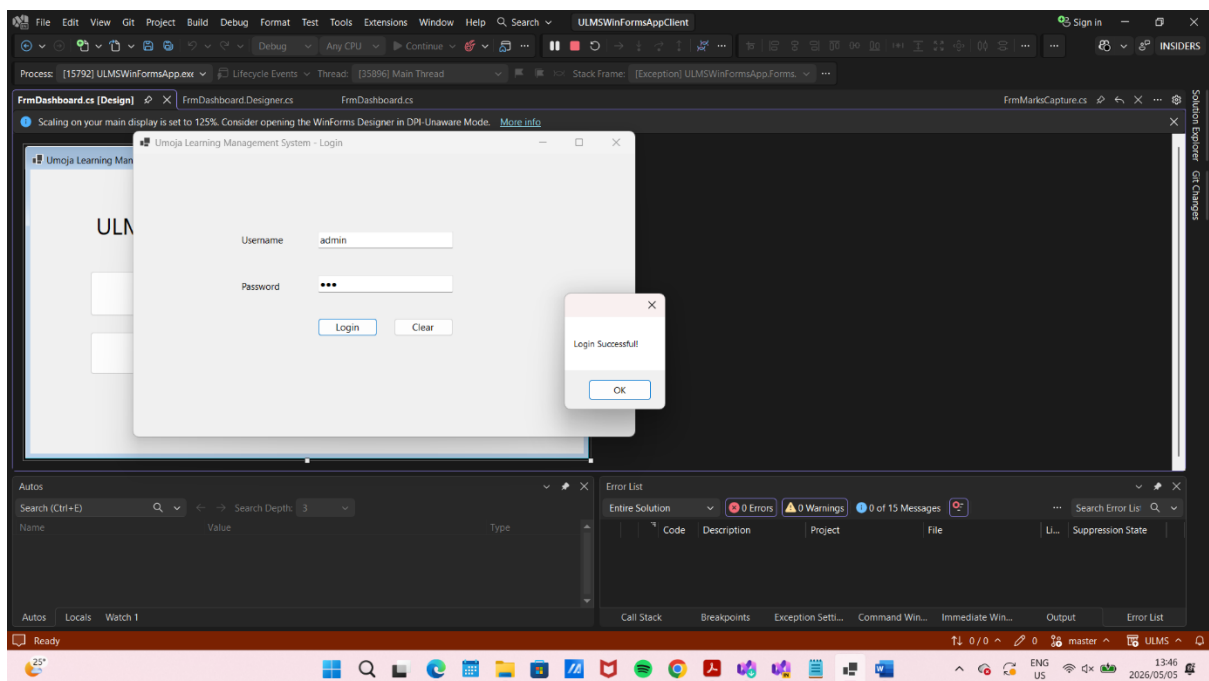
Test login with correct details

Correct username and password

Dashboard opens successfully

Login message appears stating “Login successful” and opens dashboard

Pass



Invalid login

TC002

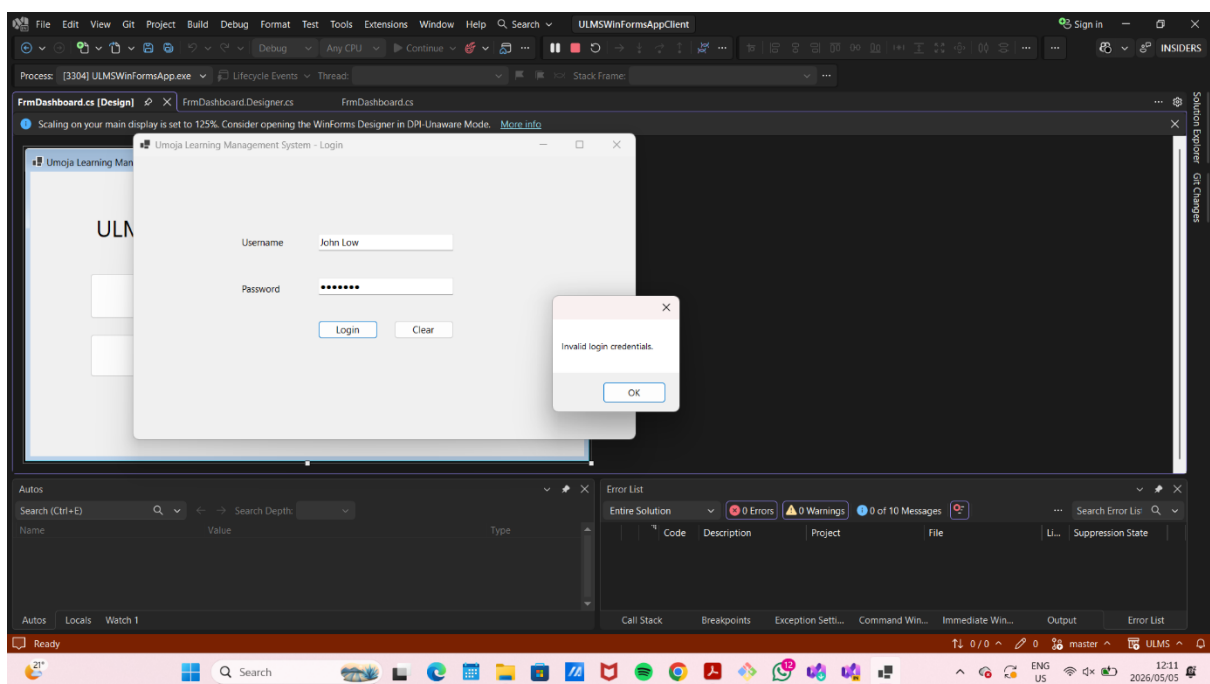
Test login with wrong password

Wrong password entered

Error message appears

Error message does appear

Pass



Empty fields

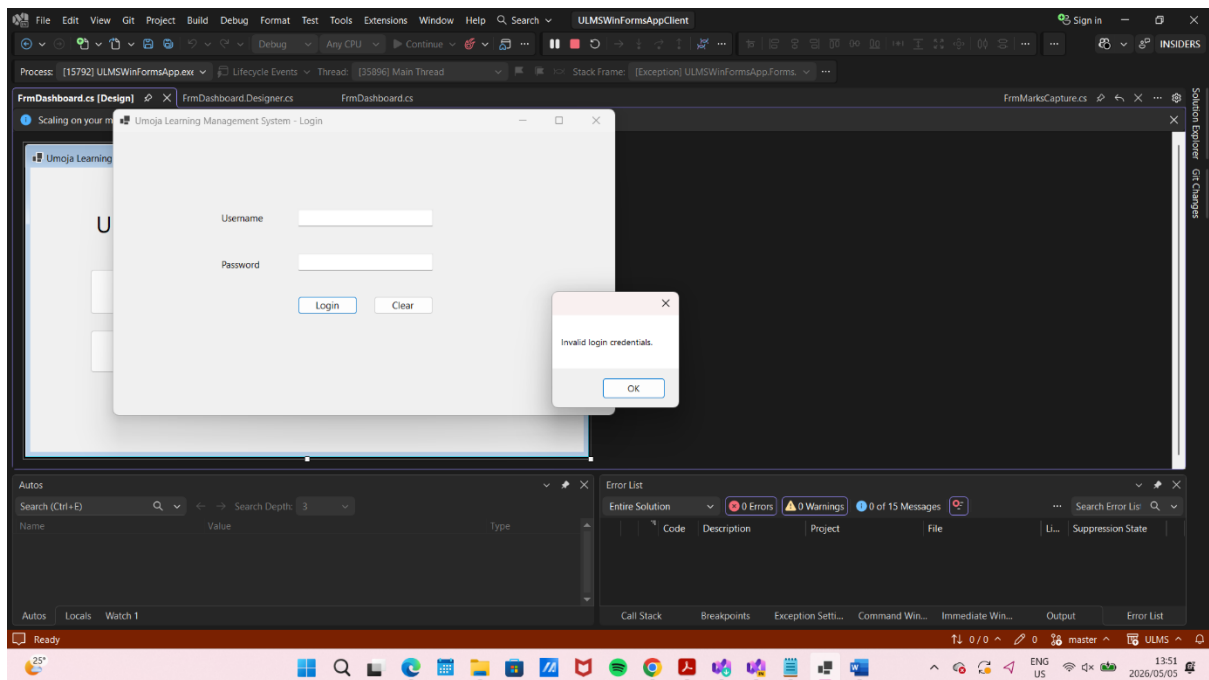
TC003

Login with empty fields

Nothing typed, clicked login

Error message appears

Pass



TC004

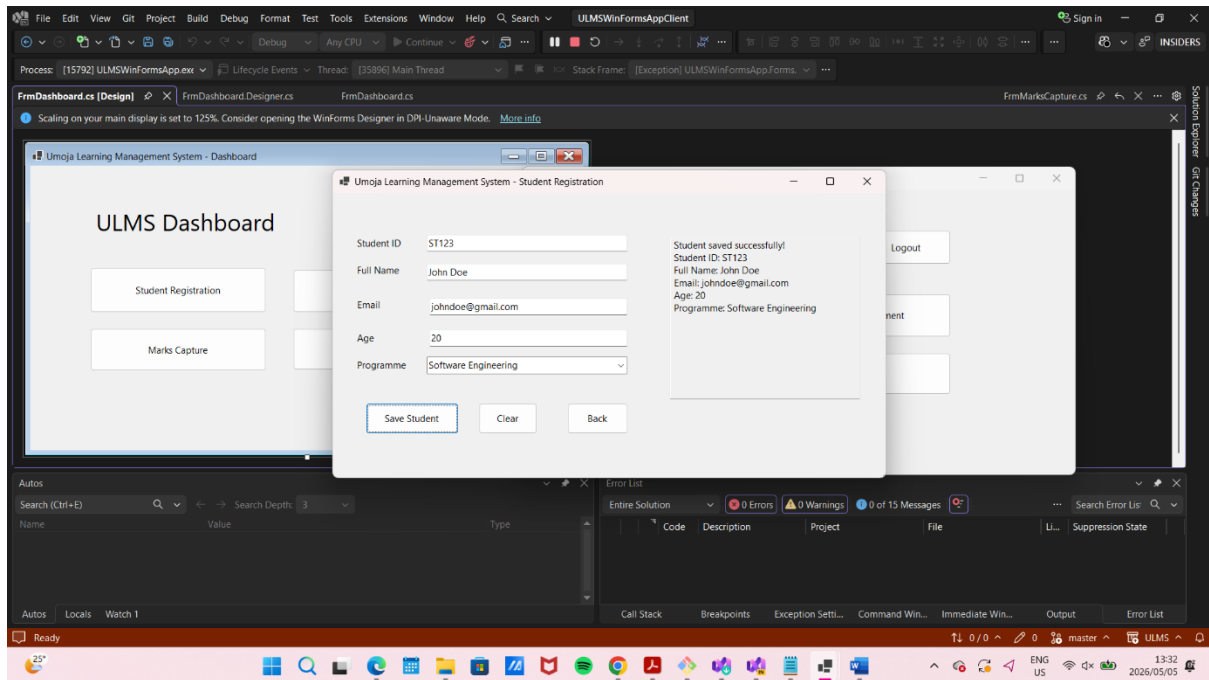
Register a new student

Valid student details entered

Student registered successfully

Student was successfully registered

Pass



Duplicate Student Registration

TC005

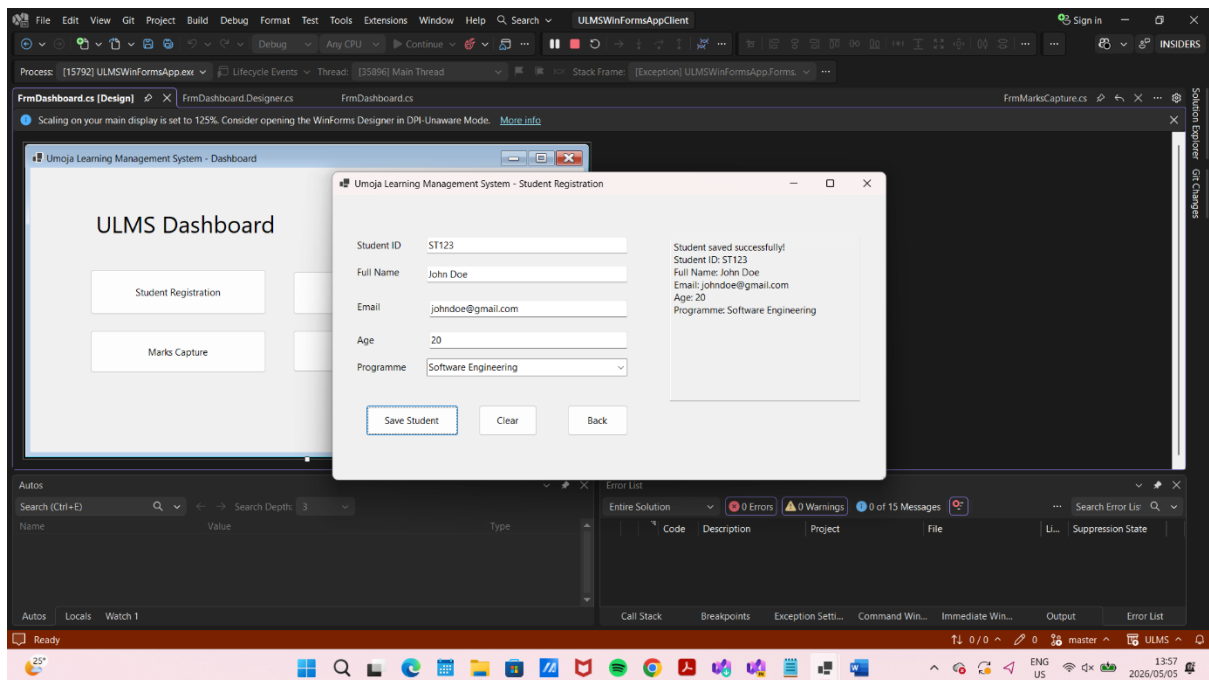
Register same student twice

Same details as TC004

Error, student already exists

Still saved the student details

Fail



Valid Mark Entry

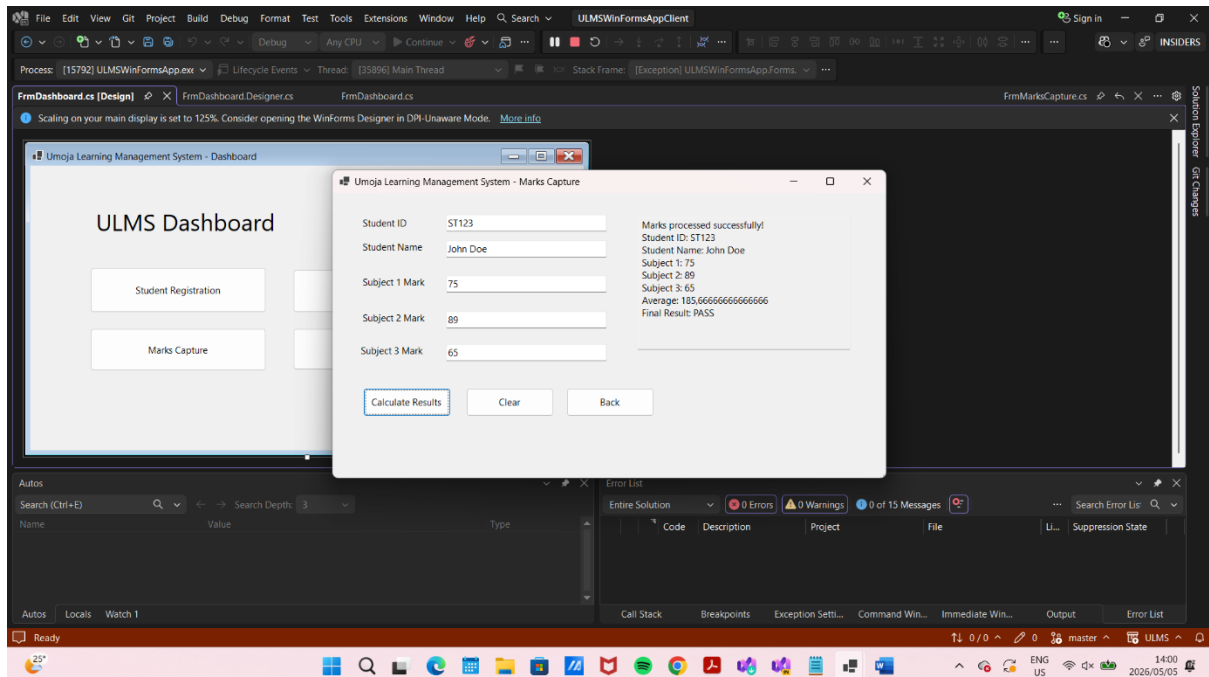
TC006

Enter valid mark

Typed 75 in mark field

Mark saved

Pass



Invalid Marks entry

TC007

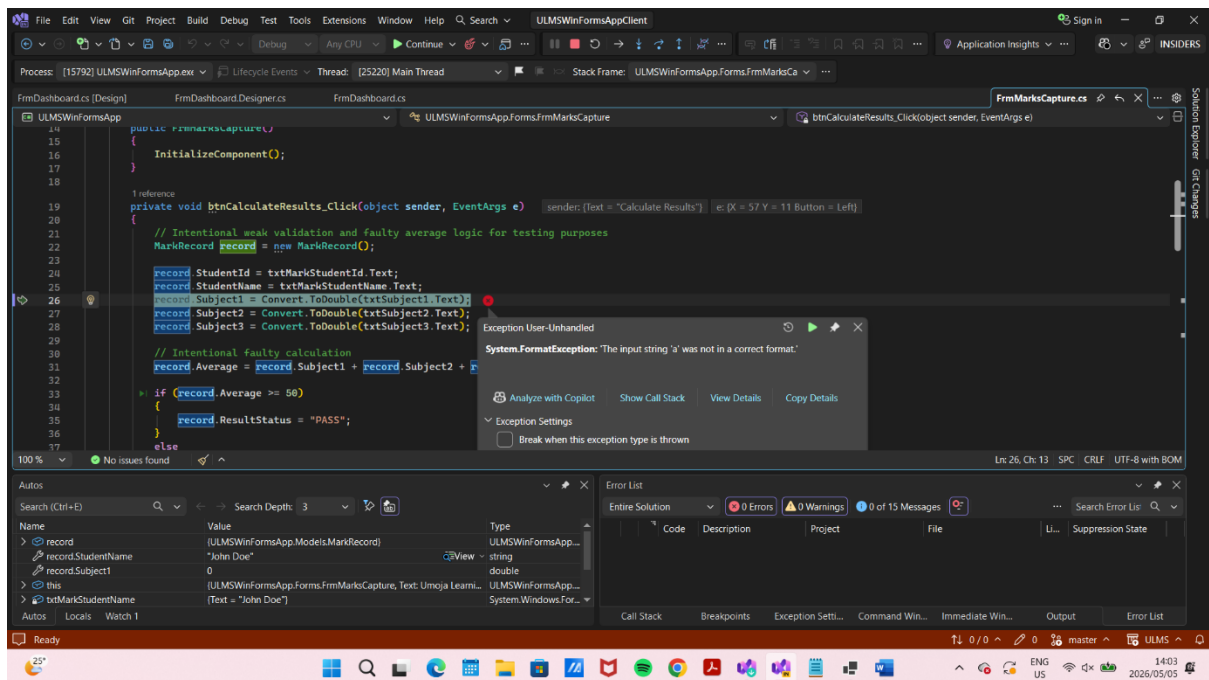
Enter letters in mark field

Typed “abc” in marks field

Error, numbers only

Does not execute the action and opens to code

Blocked



Course enrolment

TC008

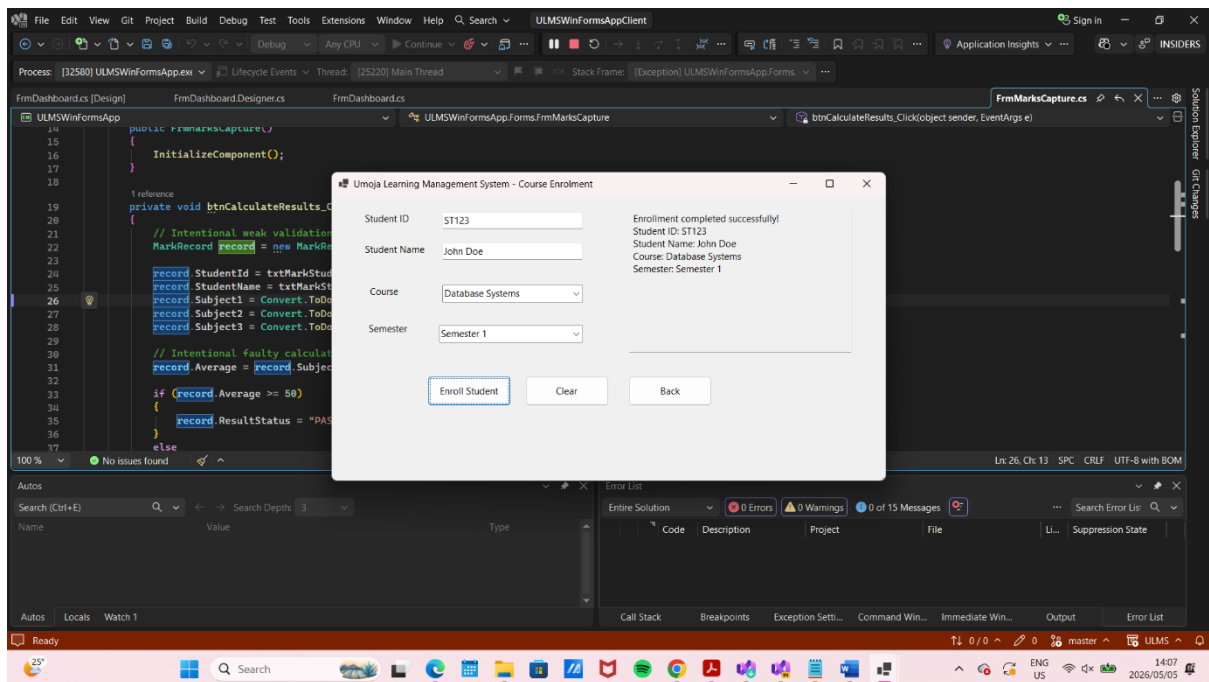
Enrol student in a course

Selected student and course

Enrolled successfully

Successfully enrolls student

Pass



Duplicate course enrolment

TC009

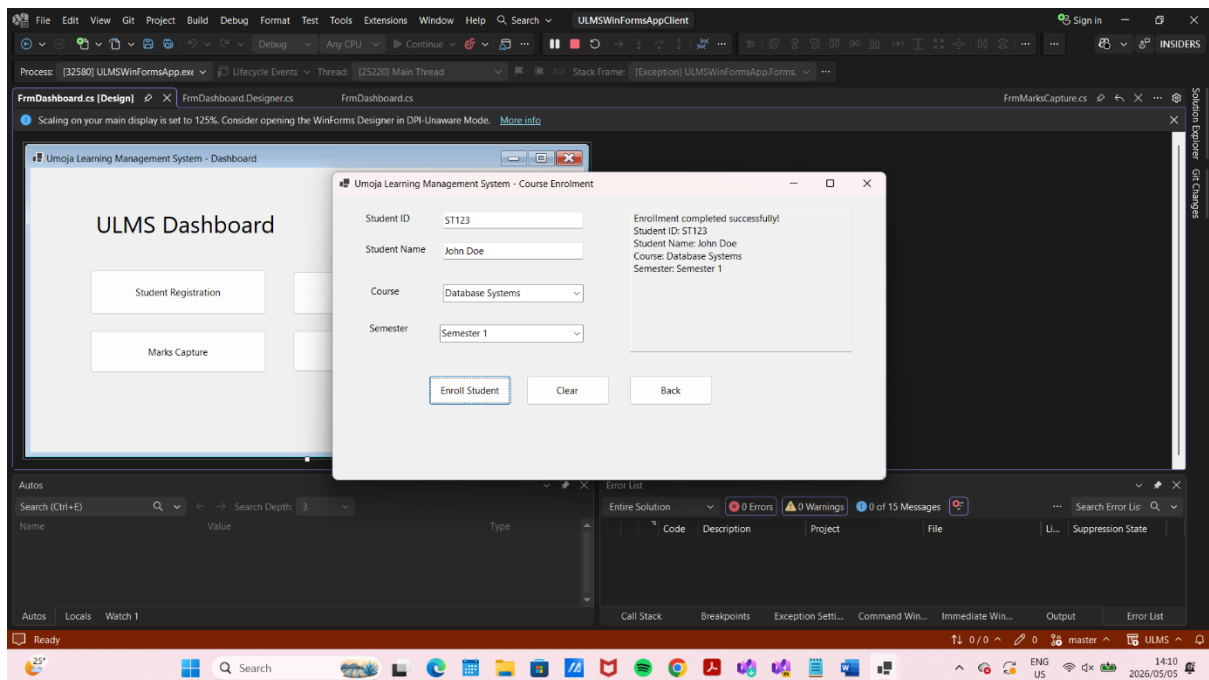
Enrol same student twice

Same student and course as TC008

Error already enrolled

Still successfully enrolls student

Fail



Report Generation Speed

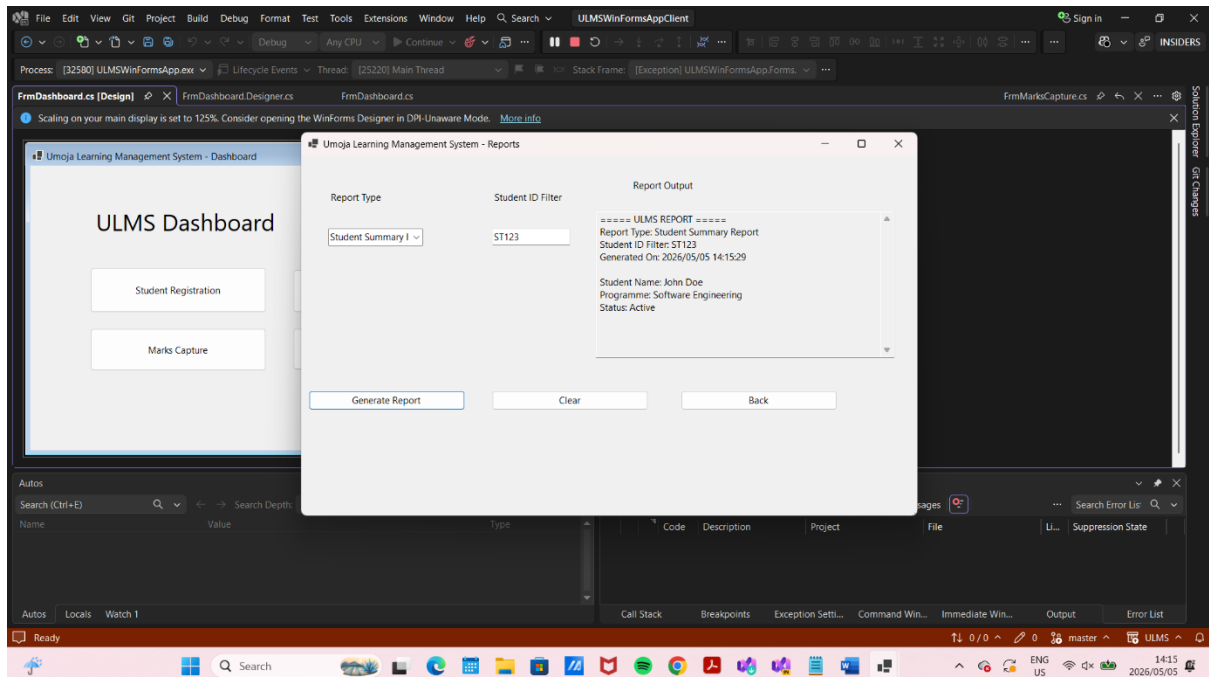
TC010

Test report loading speed

Clicked reports button

Loads within 3 seconds

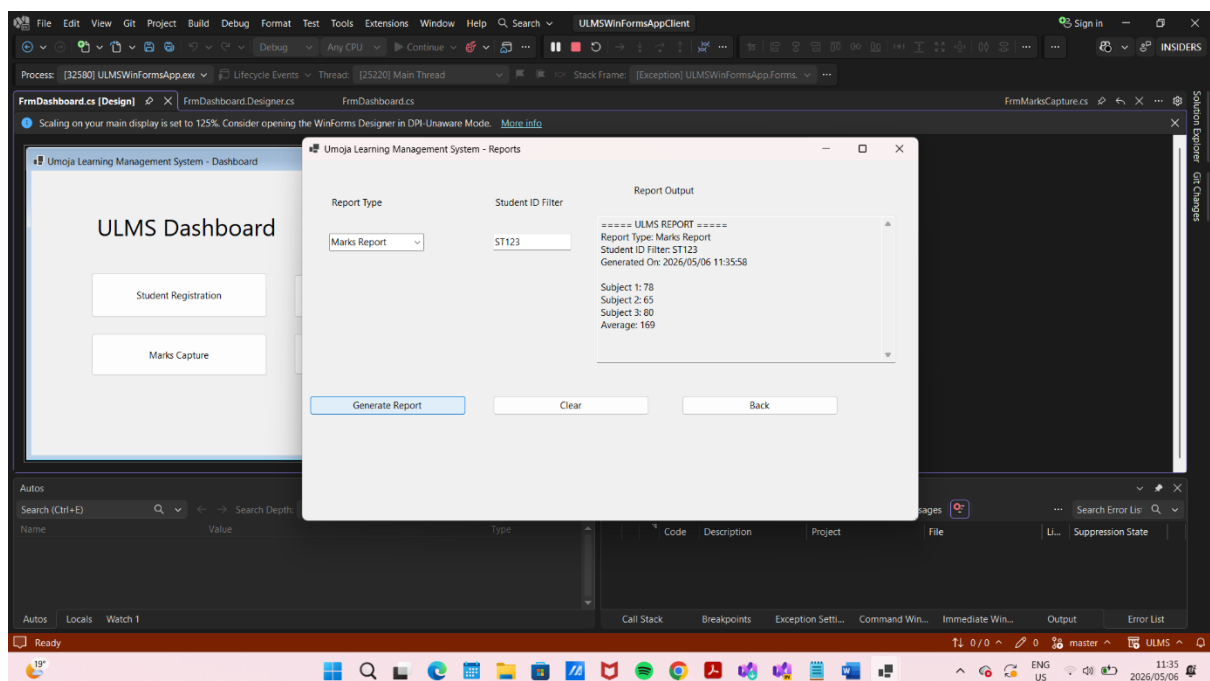
Loads slowly, takes around 5-6 seconds



Task 4

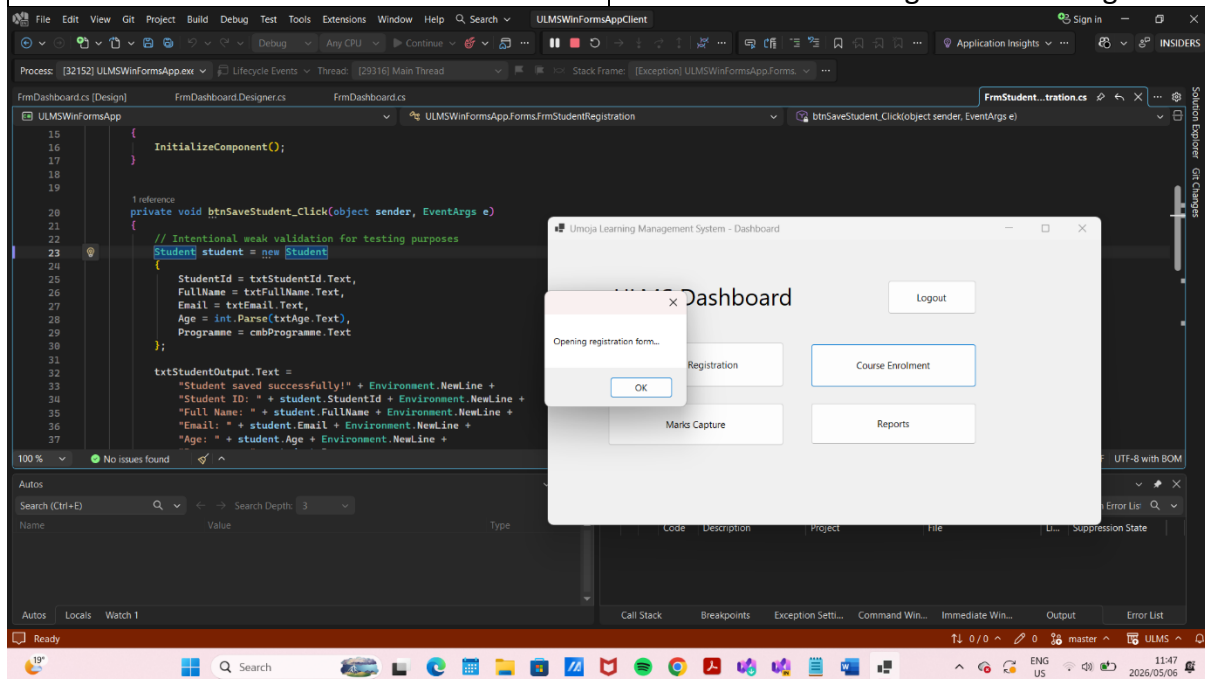
Performance Testing

Test Type:	Performance Testing
What Was Tested:	Report Loading time
Method:	Clicked reports button and timed how long it took to load
Result:	Report took 5 seconds to load
Expected:	Should load within 3 seconds
Finding:	FAIL- Report loads too slowly



Usability Testing

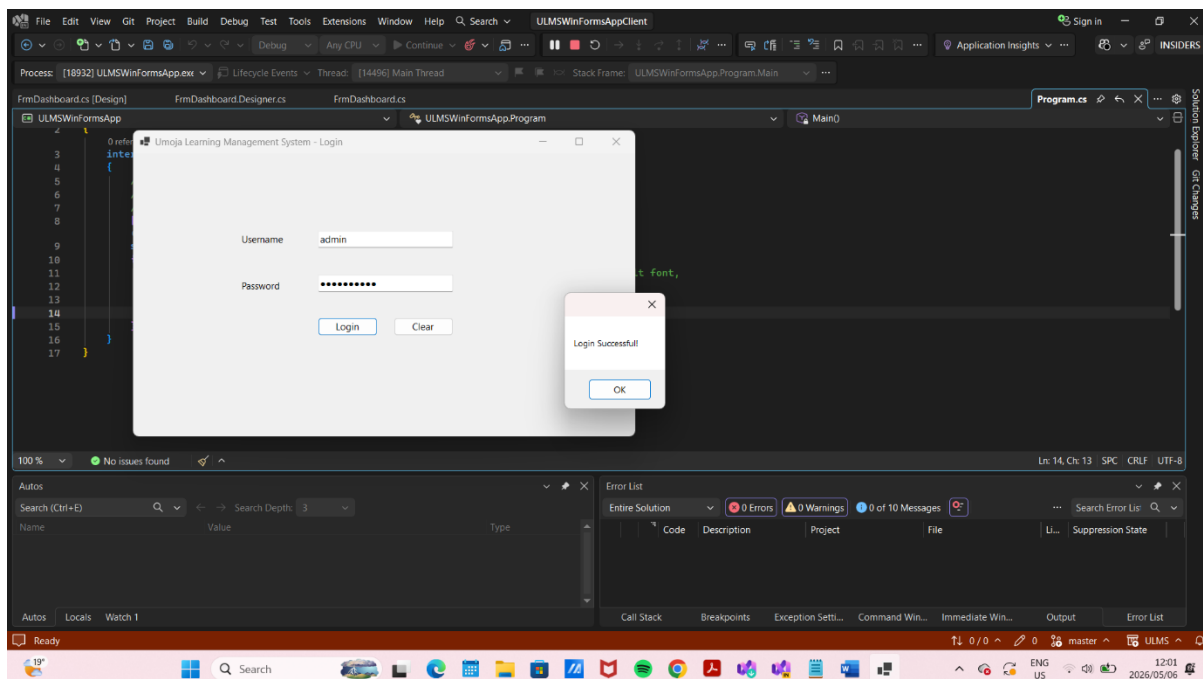
Test Type:	Usability Testing
What was tested:	Navigation between screens
Method:	Counted the number of steps needed to complete each task
Result:	
-Register a student:	3 clicks
-Enrol in course:	2 clicks
-Capture marks:	2 clicks
Finding:	Found when registering a student there is an unnecessary extra click needed after clicking the register button
Bug Found:	After completing an action, clicking anywhere reopens the Student Registration form. This makes navigation confusing



(The Course Enrolment button was the last action done, pressed on the open spaces of the program and it loaded up the Student Registration form)

Security Testing

Test Type:	Security Testing
What Was Tested:	Invalid login attempts
Method:	Attempted to login with wrong credentials 3 times
Attempt 1:	
-Input:	Wrong password
-Result:	Logged in successfully
Attempt 2:	
-Input:	Wrong password
-Result:	Logged in successfully
Attempt 3:	
-Input:	Wrong password again
-Result:	Logged in successfully
Finding:	It did not lock the account or show an error
Expected:	System should lock account or show warning after multiple failed attempts



Reliability Testing

<i>Test Type:</i>		<i>Reliability Testing</i>
<i>What Was Tested:</i>		Repeated operations
<i>Method:</i>		Performed the same actions multiple times to check
<i>Test 1 - Register student 3 times:</i>		
-Attempt 1:		Student is registered successfully
-Attempt 2:		Registered the same student
-Attempt 3:		Registered the same student again
<i>Test 1 - Course enrolment 3 times:</i>		
-Attempt 1:		-Student is enrolled successfully
-Attempt 2:		-Enrolled the same course to the same student
-Attempt 3:		-Enrolled the same course to the same student again
<i>Finding:</i>		The application behaved the same every time
<i>Expected:</i>		System should work consistently every time without crashing

Task 5

Bug 1 – Login Validation

Original faulty code:

```
15         string password = txtPassword.Text;
16
17         // Intentional faulty validation logic (for testing scenario)
18         if (username == "admin" || password == "1234")
19         {
20             MessageBox.Show("Login Successful!");
21
22             FrmDashboard dashboard = new FrmDashboard();
23             dashboard.Show();
24             this.Hide();
25         }
26         else
27         {
28             MessageBox.Show("Invalid login credentials.");
29         }
30     }
31 }
```

Issue found:

The login form does not properly validate if the username and password fields are correct. It allows either one of them to be correct for the user to be granted access.

Why It Happens:

The condition checking if anyone of the fields is written correctly so the system tries to log in, it does so successfully.

Corrected code:

The “&” is supposed to be there and not “||” to show that both the username and password have to be correct to be granted access, not one of them being correct.

```
        // Intentional faulty validation logic (for testing scenario)
        if (username == "admin" & password == "1234")
        {
            MessageBox.Show("Login Successful!");

            FrmDashboard dashboard = new FrmDashboard();
            dashboard.Show();
            this.Hide();
        }
        else
        {
            MessageBox.Show("Invalid login credentials.");
        }
    }
}
```

Bug 2 – Average Calculations

Original faulty code:

```
28 record.Subject3 = Convert.ToDouble(txtSubject3.Text);
29
30 // Intentional faulty calculation
31 record.Average = record.Subject1 + record.Subject2 + record.Subject3 / 3;
32
```

Issue Found:

The average calculation is incorrect because it divides only the last mark by 3 instead of dividing the total of all marks by 3

Why It Happens:

In coding, division is calculated before addition unless brackets are used. Without brackets around the addition only subject3 gets divided by 3

Example of the problem:

If marks are 60, 70, 80

Wrong: $60 + 70 + 80 / 3 = 156.67$

Correct: $(60 + 70 + 80) / 3 = 70$

Correct code:

```
27 record.Subject2 = Convert.ToDouble(txtSubject2.Text);
28 record.Subject3 = Convert.ToDouble(txtSubject3.Text);
29
30 // Intentional faulty calculation
31 record.Average = (record.Subject1 + record.Subject2 + record.Subject3) / 3;
32
```

Bug 3 – Slow Report Loading

Original faulty code:

```
1 reference
18 private void btnGenerateReport_Click(object sender, EventArgs e)
19 {
20     // Intentional weak validation and slow processing for testing purposes
21     string reportType = cmbReportType.Text;
22     string studentId = txtReportStudentId.Text;
23
24     // Intentional poor performance simulation
25     Thread.Sleep(4000);
26
27     StringBuilder report = new StringBuilder();
28
29     report.AppendLine("===== ULMS REPORT =====");
```

Issue found:

The code deliberately pauses for 4 seconds before loading the report. This causes the slow loading that was noticed during testing

Why It Happens:

Thread.sleep(4000) forces the application to stop and wait for 4 seconds before continuing

Corrected code:

Simply removing that line, the reports will then load immediately without delay

Bug 4 – Missing Duplicates

```
1 reference
private void btnEnroll_Click(object sender, EventArgs e)
{
    // Intentional weak business-rule validation for testing purposes
    Enrollment enrollment = new Enrollment
    {
        StudentId = txtEnrollStudentId.Text,
        StudentName = txtEnrollStudentName.Text,
        CourseName = cmbCourse.Text,
        Semester = cmbSemester.Text
    };

    txtEnrollmentOutput.Text =
        "Enrollment completed successfully!" + Environment.NewLine +
        "Student ID: " + enrollment.StudentId + Environment.NewLine +
        "Student Name: " + enrollment.StudentName + Environment.NewLine +
        "Course: " + enrollment.CourseName + Environment.NewLine +
        "Semester: " + enrollment.Semester;
}

1 reference
private void btnClearEnrollment_Click(object sender, EventArgs e)
{
    txtEnrollStudentId.Clear();
    txtEnrollStudentName.Clear();
}
```

Issue Found:

The enrolment code does not check if a student is already enrolled in the same course

before enrolling them again. It always show “Enrolment completed successfully” even for duplicates

Why It Happens:

There is no if statement checking for duplicate enrolment before saving. The code just blindly enrolls the student every time without any validation

Correct code:

```
9 // Intentional weak business-rule validation for testing purposes
9 Enrollment enrollment = new Enrollment
10 {
11     StudentId = txtEnrollStudentId.Text,
12     StudentName = txtEnrollStudentName.Text,
13     CourseName = cmbCourse.Text,
14     Semester = cmbSemester.Text
15 };
16 if (IsStudentEnrolled(enrollment.StudentId , enrollment.CourseName)){
17     MessageBox.Show("Student is already enrolled in this course!");
18     return;
19 }
20 else
21 {
22     txtEnrollmentOutput.Text =
23         "Enrollment completed successfully!" + Environment.NewLine +
24         "Student ID: " + enrollment.StudentId + Environment.NewLine +
25         "Student Name: " + enrollment.StudentName + Environment.NewLine +
26         "Course: " + enrollment.CourseName + Environment.NewLine +
27         "Semester: " + enrollment.Semester;
28 }
29 }
```

```
private partial class FrmEnrollment : Form
{
    private List<Enrollment> enrollments = new List<Enrollment>();

    // Method to check if student is already enrolled
    1 reference
    private bool IsStudentEnrolled(string studentId, string courseName)
    {
        // Check if the student is already enrolled in the same course
        return enrollments.Any(e => e.StudentId == studentId && e.CourseName == courseName);
    }
    1 reference
}
```

Task 6

Bug ID	Module	Description	Severity	Priority	Status	Resolution
BUG001	Login	Used “ ” operator instead of “&” in login validation condition	High	High	Fixed	Changed “ ” operator to “&” operator
BUG002	Reports	Thread.sleep(4000) was added to the report generation code causing the report to take 4 seconds to load every time it is loaded. This causes slow performance	High	High	Fixed	Removed the thread.sleep(4000) line from the report generation method. Reports now load immediately without delay
BUG003	Course Enrolment	The enrolment form does not check if a student is already enrolled in a course before enrolling them again. This allows the same student to be enrolled in the same course multiple times	High	High	Fixed	Added a duplicate check before saving the enrolment. If student is already enrolled an error message now displays: Student already enrolled in this course
BUG004	Dashboard	After completing any action on the dashboard , clicking anywhere on the screen reopens the student registration form unexpectedly. This affects usability and confuses users	Medium	High	Not fixed	Identified during usability testing. The form click event is incorrectly linked to the student Registration button. Needs further investigation by the development team
BUG005	Marks Capture	The marks capture module does not validate if the input is a number before calculating	High	High	Fixed	Added input validation to check that marks entered are numbers only. If

		the average. Entering letters instead of numbers causes the application to crash with no helpful error message shown to the user				letters are entered an error message now displays: "Please enter numbers only"
--	--	---	--	--	--	--

Umoja Learning Management System

Prepared by : Shatadi Nkadimeng

Student Number: 20240842

Executive Summary

This report presents the finding of the software testing activities performed on the Umoja Learning Management System. The system was tested by Junior Software Tester before production deployment. The testing was conducted across all major features of the system including student registration, login authentication, course enrolment, marks capture and report generation. A total of 10 test cases were designed and executed. 5 defects were identified during testing, 4 of the 5 defects have been successfully fixed. 1 defect remains open and requires further investigation by the development team

Testing Scope

Module	Tested
Student Registration	Yes
Login Authentication	Yes
Course Enrolment	Yes
Marks Capture	Yes
Report Generation	Yes

The following testing types were performed:

- Functional Testing
- Input Validation Testing
- Integration Testing
- Performance Testing
- Usability Testing
- Security Testing
- Reliability Testing

Tools used

Tools	Purpose
Visual Studio	Running and debugging code
Github	Cloning the project
Windows Snipping Tool	Taking screenshots as evidence
Microsoft Word	Documenting test results
Manual Testing	Executing test cases
Phone timer	Measuring report load time

Functional Testing results:

Total test cases executed: 10

Passed: 7

Failed: 2

Blocked: 1

Key Findings:

- Login validation contains a coding error using “|” operator instead of “&” operator
- Marks capture does not validate letter input
- Course enrolment allows duplicate enrolment with not error message
- Student Registration form reopens unexpectedly after any action

Non-functional testing results:

Performance testing:

- Report generation took over 4 seconds to load due to `thread.sleep(4000)` found in the code
- Expected load time is under 3 seconds
- Result: FAIL

Usability testing:

- Navigation between screens works however a bug causes the Student Registration form to reopen when clicking anywhere on the dashboard
- Result: FAIL

Security Testing:

- System does not lock account after multiple failed login attempts
- Only one hardcoded username and password exists

- Result: FAIL

Reliability testing:

- System remained stable during repeated operations
- No crashes were recorded during repeated registrations and enrolment
- Result: PASS

Defects Identified

<i>Bug ID</i>	<i>Module</i>	<i>Severity</i>	<i>Status</i>
BUG001	Login	High	Fixed
BUG002	Reports	High	Fixed
BUG003	Enrolment	High	Fixed
BUG004	Dashboard	Medium	Not Fixed
BUG005	Marks	High	Fixed

Total defects found: 5

Total fixed: 4

Total not fixed: 1

Corrections Implemented

Correction 1- Login Validation:

Changed “||” operator to “&” operator in the login condition to ensure correct logical comparison behaviour

Correction 2- Report Performance:

Removed thread.sleep(4000) from the report generation method. Reports now load immediately without artificial delay

Correction 3- Duplicate Enrolment:

Added a duplicate check before saving enrolments. System now shows an error message if student is already enrolled in the selected course

Correction 4 – Mark Validation:

Added input validation to marks capture to ensure only numbers are accepted. System now shows an error message if letters are entered

Recommendations

- 1) Fix the dashboard bug where student registration form reopens unexpectedly after any action is completed
- 2) Improve security by adding account lockout after 3 failed login attempts
- 3) Replace hardcoded login credentials with a proper database driven authentication system
- 4) Add proper exception handling across all modules to prevent the app from crashing without helpful error message
- 5) Conduct further regression testing after all bugs have been fixed to ensure no new bugs were introduced

Deployment Readiness Decision

Decision: Not ready for deployment

Reason:

Although 4 out of 5 defects have been fixed, the system is not yet ready for production deployment for the following reason:

- 1) The dashboard navigation bug has not been resolved and will confuse users
- 2) Security vulnerabilities exist, the system uses hardcoded credentials and does not lock account after failed login attempts
- 3) Further regression testing is required after all fixes are implemented to confirm the system is stable

Recommendations:

Fix all remaining defects, conduct a final round of testing and resubmit for deployment approval

Github link: <https://github.com/Shataadi/Formal-Assessment-2>